The eigenproblem

For $A \in \mathbb{C}^{n \times n}$, suppose a scalar λ and nonzero vector x satisfy

$$Ax = \lambda x,$$

Then λ is an eigenvalue and x an eigenvector.

Equivalently,

$$(A - \lambda I)x = 0.$$

Since $x \neq 0$, the matrix $A - \lambda I$ is singular. Hence

$$p_A(\lambda) = \det(\lambda I - A) = 0.$$

The one-dimensional subspace $\operatorname{span}\{x\}$ is invariant. Over the reals, a negative eigenvalue reverses the arrow direction while preserving the line.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Characteristic polynomial, trace, and determinant

Using the monic convention for an $n \times n$ matrix,

$$p_A(\lambda) = \lambda^n - a_1\lambda^{n-1} + \cdots + (-1)^n \det A.$$

Counting eigenvalues with algebraic multiplicity gives

$$\sum_i \lambda_i = \operatorname{tr} A, \qquad \prod_i \lambda_i = \det A.$$

For example,

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

The trace is 6 and determinant is 8.

$$p_A(\lambda) = \lambda^2 - 6\lambda + 8 = (\lambda - 4)(\lambda - 2).$$

The eigenvalues are 4 and 2. Trace and determinant check the spectrum but do not generally determine it in dimensions above two.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

How operations transform eigenvalues

If $Ax = \lambda x$, then for nonnegative integers k and polynomials q,

$$A^k x = \lambda^k x,$$

$$A^{-1}x = \lambda^{-1}x \quad (A \text{ invertible}),$$

$$q(A)x = q(\lambda)x.$$

Shift:

$$(A + cI)x = (\lambda + c)x.$$

Eigenspace:

$$E_\lambda = N(A - \lambda I).$$

The eigenspace is a subspace whose dimension is the geometric multiplicity. A triangular matrix has its diagonal entries as eigenvalues, counted with algebraic multiplicity.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Algebraic and geometric multiplicities

The multiplicity of a root in the characteristic polynomial is its algebraic multiplicity a_λ. The quantity

$$g_\lambda = \dim N(A - \lambda I)$$

is its geometric multiplicity. Extend an eigenspace basis to a basis of the whole space. The resulting matrix is block upper triangular with block λI_{g_λ}, so its characteristic polynomial contains $(t - \lambda)^{g_\lambda}$. Therefore

$$1 \le g_\lambda \le a_\lambda.$$

If any eigenvalue has $g_\lambda < a_\lambda$, the matrix is defective.

For example,

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}.$$

The algebraic multiplicity is 2. Solving

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Algebraic and geometric multiplicities (continued)

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x = 0$$

gives only $x_2 = 0$, hence

$$E_2 = \text{span}\{(1,0)^T\}, \qquad g_2 = 1.$$

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

A complete two-dimensional example

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

$\lambda = 4$:

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} x = 0,$$

Choose $x_1 = (1,1)^T$.

$\lambda = 2$:

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} x = 0,$$

Choose $x_2 = (1,-1)^T$.

These eigenvectors are orthogonal because A is symmetric. If

$$u_0 = c_1 x_1 + c_2 x_2,$$

Then

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

A complete two-dimensional example (continued)

$$u_k = A^k u_0 = c_1 4^k x_1 + c_2 2^k x_2.$$

For $c_1 \neq 0$, the eigenvalue-4 mode determines the limiting direction.

Taking $c_1 = c_2 = 1$,

$$u_0 = x_1 + x_2 = (2, 0)^T, \quad u_1 = (6, 2)^T, \quad u_2 = (20, 12)^T.$$

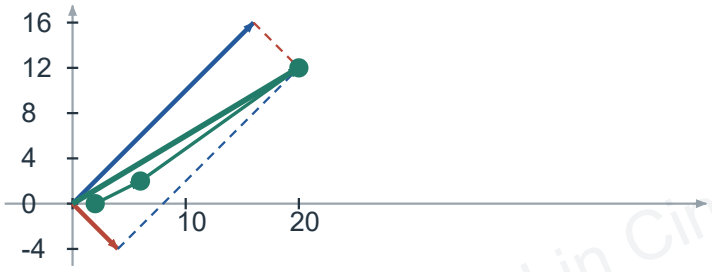
The diagram shows $c_1 4^k x_1$, $c_2 2^k x_2$, and their sum u_k. For $c_1 = 0, c_2 = 1$, instead,

$$u_0 = (1, -1)^T, \quad u_1 = (2, -2)^T, \quad u_2 = (4, -4)^T,$$

the trajectory stays in the eigenvector direction x_2. It has no component in the larger-eigenvalue mode. Real coefficients can also be used with unit eigenvectors; here the diagram retains the specified nonunit x_1, x_2, so its coefficients refer to that basis.

A complete two-dimensional example (continued)

Eigenmodes: components, sum, and trajectory



A=[[3,1],[1,3]]

$\lambda_1=4 \cdot x_1=(1,1)$

$\lambda_2=2 \cdot x_2=(1,-1)$

Blue: $4^2 x_1=(16,16)$

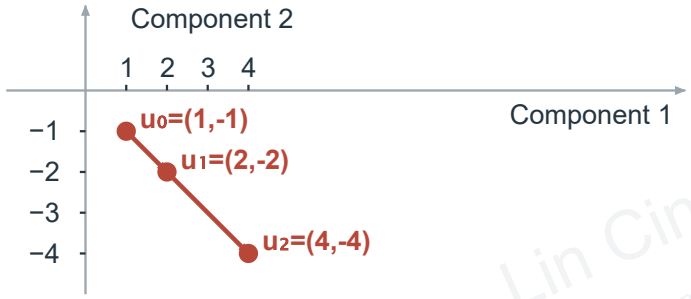
Red: $2^2 x_2=(4,-4)$

Green: $u_2=(20,12)$

$u_0,u_1,u_2=(2,0),(6,2),(20,12)$

A complete two-dimensional example (continued)

Exceptional trajectory $c_1=0$: always along $x_2=(1,-1)$



$c_1=0$ · $c_2=1$

$u_k=2^k(1,-1)$

Every u_k has the same eigenvector direction.

Fixed ratio; no 4^k dominant term.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Rotation and complex eigenvalues

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

$$p(\lambda) = \lambda^2 - 2\cos\theta\,\lambda + 1,$$

$$\lambda = e^{\pm i\theta}.$$

Unless $\theta \equiv 0 \pmod{\pi}$, a planar rotation has no nonzero real eigenvector.

Complex eigenvalues of a real matrix come in conjugate pairs. If

$$Ax = \lambda x,$$

taking complex conjugates gives

$$A\overline{x} = \overline{\lambda}\,\overline{x}.$$

Real solutions require compatible conjugate coefficients. A single complex mode is generally not a real physical solution.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Diagonalization

Suppose A has n independent eigenvectors x_i. Form

$$X = [x_1 \ \cdots \ x_n].$$

$$AX = X\Lambda, \qquad \Lambda = \operatorname{diag}(\lambda_i),$$

$$A = X\Lambda X^{-1}, \qquad X^{-1}AX = \Lambda.$$

Diagonalization is equivalent to the eigenspace dimensions summing to n. Eigenvectors for distinct eigenvalues are independent. Indeed, a shortest nontrivial relation $\sum c_i x_i = 0$, after applying $A - \lambda_k I$, would give a shorter relation, a contradiction. Thus n distinct eigenvalues suffice. Repeated eigenvalues can also have complete eigenspaces, as for $2I$.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Powers and difference equations

$$A^k = X\Lambda^k X^{-1}.$$

Recurrence

$$u_{k+1} = Au_k$$

has solution

$$u_k = \sum_{j=1}^{n} c_j \lambda_j^k x_j.$$

If every

$$|\lambda_j| < 1,$$

zero solution asymptotically stable.Spectral radius:

$$\rho(A) = \max_j |\lambda_j|.$$

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Powers and difference equations (continued)

then every fixed initial vector decays to zero. The condition $\rho(A) < 1$ guarantees this even for defective matrices, since polynomial factors cannot overcome geometric decay. Eigenvalues on the unit circle can give bounded, periodic, or defective polynomial growth.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Similarity invariants

If

$$C = B^{-1}AB,$$

then

$$C - \lambda I = B^{-1}(A - \lambda I)B,$$

Thus characteristic polynomial, eigenvalues, trace, determinant, rank, and Jordan structure are unchanged.

If $Ax = \lambda x$, set $y = B^{-1}x$. Then

$$Cy = \lambda y.$$

Similarity changes coordinates while preserving the intrinsic spectrum.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Defective matrices and Jordan structure

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

There is only one eigendirection, so no similarity can diagonalize this matrix.

Write

$$A = 2I + N, \qquad N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad N^2 = 0.$$

Since $2I$ and N commute,

$$A^k = 2^k I + k2^{k-1} N = \begin{bmatrix} 2^k & k2^{k-1} \\ 0 & 2^k \end{bmatrix}.$$

Let $e_1 = (1, 0)^T, e_2 = (0, 1)^T$. Then

$$(A - 2I)e_1 = 0, \qquad (A - 2I)e_2 = e_1.$$

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Defective matrices and Jordan structure (continued)

Thus e_1 is an eigenvector, e_2 a generalized eigenvector, and (e_1, e_2) a Jordan chain of length 2. The diagonal describes scaling along e_1; the off-diagonal mixes e_1 into the image of e_2, producing the factor $k2^{k-1}$ in matrix powers. This is the Jordan block $J_2(2)$. Its missing independent eigenvector prevents a decomposition of the whole space into eigenvector lines.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Spectral decomposition

If the rows of X^{-1} are y_j^T,

$$A = \sum_{j=1}^{n} \lambda_j x_j y_j^T.$$

Each $x_j y_j^T$ has rank one. For a real symmetric matrix choose a real orthogonal matrix Q:

$$A = Q \Lambda Q^T = \sum_j \lambda_j q_j q_j^T.$$

This scales independent orthonormal directions. Nonnormal matrices generally have nonorthogonal eigenvectors, so eigenvalues alone do not describe transient amplification.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

From scalar to matrix ODEs

Scalar:

$$u' = \lambda u \quad \Longrightarrow \quad u(t) = Ce^{\lambda t}.$$

System:

$$u' = Au, \qquad u(0) = u_0.$$

If A is diagonalizable,

$$u(t) = \sum_j c_j e^{\lambda_j t} x_j.$$

$$c = X^{-1} u_0.$$

For $\lambda = \alpha + i\beta$,

$$e^{\lambda t} = e^{\alpha t}(\cos \beta t + i \sin \beta t).$$

The real part determines exponential growth or decay. The imaginary part determines oscillation.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

The matrix exponential

$$e^{At} = I + At + \frac{A^2t^2}{2!} + \cdots.$$

The absolutely convergent power series can be differentiated term by term on bounded time intervals.

$$\frac{d}{dt}e^{At} = Ae^{At} = e^{At}A.$$

Diagonalizable:

$$e^{At} = Xe^{\Lambda t}X^{-1} = \sum_j e^{\lambda_j t}x_j y_j^T.$$

Defective $A = 2I + N$, $N^2 = 0$:

$$e^{At} = e^{2t}(I + tN).$$

Eigenvalue real parts determine asymptotic stability. Nonnormal transient behavior also depends on norms or more detailed information about the matrix.

Stability and its boundary cases

$$\begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}.$$

Use eigenbasis：

$$u(t) = c_1 e^{4t}(1,1)^T + c_2 e^{2t}(1,-1)^T.$$

Both eigenvalues are positive, so every nonzero initial vector produces a growing solution.

The zero solution of a linear system is asymptotically stable exactly when every eigenvalue has negative real part. A Jordan block $J = \lambda I + N$ of size m gives $e^{Jt} = e^{\lambda t}\sum_{\ell=0}^{m-1} t^\ell N^\ell/\ell!$. A negative real part makes every polynomial times the decaying exponential tend to zero. A positive real part gives an unstable mode.

Stability and its boundary cases (continued)

At the boundary, if all real parts are nonpositive and every imaginary-axis Jordan block has size one, e^{At} is bounded and the zero solution is Lyapunov stable. Any remaining imaginary-axis mode prevents asymptotic stability. For $A = 0$, solutions retain their initial values; for skew-symmetric rotation, they retain their length. In contrast, $J = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ gives $e^{Jt} = I + tJ$, taking e_2 to $(t, 1)^T$. Zero real parts can therefore still be unstable. These conclusions concern the stated linear system.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Converting a second-order ODE

$$u'' + Bu' + Cu = 0.$$

For scalar constants B, C, set $v = u'$.

$$\frac{d}{dt}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -C & -B \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix}.$$

Characteristic polynomial:

$$\det\begin{bmatrix} -\lambda & 1 \\ -C & -B-\lambda \end{bmatrix} = \lambda^2 + B\lambda + C.$$

Complex conjugate eigenvalues produce oscillation and their real parts determine decay. The initial data $(u(0), u'(0))$ become a two-dimensional initial vector.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Real eigenvalues of real symmetric matrices

Let $S \in \mathbb{R}^{n \times n}$, $S = S^T$, and choose a nonzero $x \in \mathbb{C}^n$ with $Sx = \lambda x$. Real entries and symmetry give $S^* = \overline{S}^T = S$. Using the conjugate transpose,

$$x^* S x = \lambda x^* x.$$

because

$$\overline{x^* S x} = x^* S^* x = x^* S x,$$

the numerator is real. Since $x^* x > 0$,

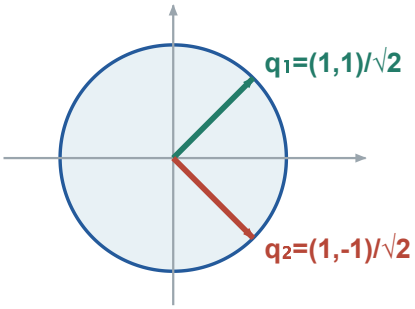
$$\lambda = \frac{x^* S x}{x^* x} \in \mathbb{R}.$$

The first equality uses conjugation of a scalar; the second uses $S^* = S$, which does not follow from $S^T = S$ for a complex matrix. The denominator $x^* x = \sum_j |x_j|^2 > 0$ establishes the conclusion.

The counterexample $S = [i]$ satisfies $S^T = S$ but $S^* = [-i] \neq S$, with eigenvalue i. The complex-space premise is Hermitian symmetry $S^* = S$. The same proof gives real eigenvalues, and diagonalization uses a unitary matrix $S = U \Lambda U^*$.

Real eigenvalues of real symmetric matrices (continued)

Orthonormal eigenbasis: $Q^T S Q = \mathrm{diag}(4,2)$



S=[[3,1],[1,3]]

Q=(1/√2)[[1,1],[1,-1]]

$Q^T S Q = \mathrm{diag}(4,2)$

$S = Q \Lambda Q^T$; the axes are perpendicular.

Normalize (1,1),(1,-1) before forming Q.

Real eigenvalues of real symmetric matrices (continued)

Complex symmetric does not imply Hermitian

S=[i] , S^T=S

Only eigenvalue: i

S*=[-i]≠S

The Hermitian premise is necessary.

Hermitian : S*=S

x*Sx=λx*x

x*x>0 ⇒ λ∈R

S=UΛU*, with U unitary

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Orthogonality of distinct eigenspaces

Let $S \in \mathbb{R}^{n\times n}$ be real symmetric, with real eigenvectors x, y. If $Sx = \lambda x$, $Sy = \mu y$,

$$\lambda x^T y = (Sx)^T y = x^T S^T y = \mu x^T y.$$

Consequently,

$$(\lambda - \mu) x^T y = 0.$$

then $\lambda \neq \mu$ implies $x \perp y$. A complete orthonormal eigenbasis follows by induction. The continuous quadratic form $x^T S x$ attains a maximum on the unit sphere; its extremum condition gives a real unit eigenvector q. For $v \perp q$, $q^T S v = (Sq)^T v = 0$, so $q^\perp$ is invariant. The restriction remains symmetric. Induction gives a complete orthonormal eigenbasis; within repeated eigenspaces, use Gram–Schmidt.

Spectral theorem：

$$S = Q \Lambda Q^T.$$

For example,

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Orthogonality of distinct eigenspaces (continued)

$$S = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}, \qquad Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Since $Q^T Q = I$ and

$$Q^T S Q = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix},$$

we obtain $S = Q \Lambda Q^T$. The unnormalized vectors $(1, 1)^T$, $(1, -1)^T$ specify directions only. Divide them by $\sqrt{2}$ before using them as columns of Q.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Equivalent positive-definiteness conditions

For real symmetric S, the following conditions are equivalent.

1. $x^T S x > 0$ for every $x \neq 0$.
2. Every eigenvalue is positive.
3. Every leading principal minor is positive.
4. Symmetric elimination without swaps has positive pivots.
5. $S = A^T A$ for some full-column-rank A.
6. $S = R R^T$ for an invertible lower-triangular R with positive diagonal, the Cholesky factorization.

Symmetric elimination completes squares, giving $S = L D L^T$ and $x^T S x = (L^T x)^T D (L^T x)$. Positive pivots are therefore equivalent to positive energy. Leading minors satisfy $\Delta_k = d_1 \cdots d_k$, so positivity of all leading minors is equivalent to positive pivots. Taking $R = L D^{1/2}$ gives Cholesky, and $A = R^T$ gives the Gram form. Conversely, squared lengths give positive energy. A positive determinant alone is insufficient; $-I_2$ has determinant 1 but is negative definite.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Equivalent positive-definiteness conditions (continued)

2×2：

$$S = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

is positive definite exactly when $a > 0, ac - b^2 > 0$.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Energy proof and counterexamples

$$x^T S x = \sum_i \lambda_i (q_i^T x)^2.$$

If all $\lambda_i > 0$, every nonzero x has some $q_i^T x \neq 0$, making the energy positive. Conversely, an eigenvalue $\lambda_i \leq 0$ supplies a nonzero eigenvector with nonpositive energy, contradicting strict positive definiteness.

For a counterexample,

$$S_1 = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

eigenvalues $3, -1$，not positive definite。

If A has independent columns,

$$x^T A^T A x = \|Ax\|^2 > 0$$

for every $x \neq 0$, so $A^T A$ is positive definite. Full row rank cannot replace column independence.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Positive semidefinite boundary

$$S = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

At $x = (1, -1)^T$, $x^T S x = 0$; the eigenvalues are $2, 0$. Thus the matrix is positive semidefinite but not positive definite.

For $S = A^T A$, dependent columns give nonzero x with $Ax = 0$ and zero energy. Conversely, zero energy forces $Ax = 0$.

A semidefinite matrix can have an LDL^T factorization, but Cholesky must handle zero pivots. A pseudoinverse does not make a semidefinite matrix invertible.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Ellipsoids and conditioning

Positive definite S:

$$x^T S x = 1$$

is an origin-centered ellipsoid. In eigen-coordinates,

$$y = Q^T x,$$
$$\sum_i \lambda_i y_i^2 = 1.$$

The semiaxis length in direction i is

$$1/\sqrt{\lambda_i}.$$

The smallest eigenvalue gives the longest semiaxis. Also,

$$\kappa_2(S) = \frac{\lambda_{\max}}{\lambda_{\min}}.$$

Positive definiteness guarantees exact invertibility, but a small minimum eigenvalue relative to the largest still gives poor conditioning.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Chapter check

You can now do the following.

- Find eigenvalues from the characteristic equation.
- Distinguish algebraic and geometric multiplicities.
- Give defective counterexamples.
- Solve powers and ODEs by diagonalization when an eigenbasis exists.
- Combine complex conjugate modes into real solutions.
- Prove the real symmetric spectral theorem and orthogonality relations.
- Connect energy, minors, pivots, Cholesky, and Gram factorizations.
- Interpret conditioning through ellipsoid semiaxes.

SVD in Chapter 7 applies to rectangular matrices and does not require a complete eigenbasis of the original matrix.

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study